



BAXI 2213 / BITI 2213: KNOWLEDGE BASED SYSTEM

SEMESTER 1 2025/2026

PROJECT (10%)

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INTRODUCTION

The title of project is Intelligent Career Path Navigator. The system operates within the Information Technology (IT) sector, focusing on career counselling for various levels of education, including Diploma, Degree, Masters, and PhD. The purpose of this system is to help the users discover a personalized IT career path by identifying their personalities and provide a learning roadmap with Malaysia salary insights.

PROBLEM BACKGROUND

In this era, navigating the IT industry is difficult due to the variety of specialized roles such as DevOps, AI, Cybersecurity and others. Users often lack a roadmap for the specific skills and certifications required for their personality and technical level. This system needs a decision-support knowledge-based system because it uses rule-based logic to correlate technical skills such as programming, math, analytics with personality traits like a stress handling, leadership, innovation to provide a tailored "Match Score" that a generic search engine cannot offer.

OBJECTIVE

- To categorize users into one of six personality archetypes (e.g., Architect, Innovator, Strategist) based on work-style traits.
- To recommend specific IT career paths with a confidence level (Very High to Medium) and a match percentage.
- To generate a structured 3-phase learning roadmap including timeframes, certifications, and project ideas.

DATA COLLECTION & FACTS

The Intelligent Career Path Navigator's knowledge base was assembled from several sources. First, subject experts such as career counselors, HR recruiters, and IT professionals shared their perspectives on career options in the IT sector. Second, current wage statistics and job demand in Malaysia were obtained from online employment portals like LinkedIn and JobStreet Malaysia. Third, industry assessments on IT job trends included the MDEC Digital Economy Report and the World Economic Forum Future of Jobs Report 2023. Fourth, the development of the knowledge base was aided by technical references like the O*NET Career Database and CLIPS Documentation. Finally, the learning roadmaps were developed using data from certification organizations such as CompTIA, AWS, and Google Career Certificates.

The facts and variable use to run the system's inference engine are categorized into four key profiles:

- **Basic User Information:** the system will record the user's name, age, level of education (from diploma to PhD), and current employment status.
- **Technical Profile:** This evaluates the user's foundation based on factors including programming ability (from expert to none), mathematical competence, analytical thinking strength, practical coding experience, and perceived learning speed.
- **Personality Profile:** The KBS assesses soft skills and preferred work styles, such as logical vs creative approaches to problem-solving, communication skills, attention to detail, leadership experience, stress tolerance, and an innovative mentality.
- **Specific Career Interests:** The algorithm determines the user's preference for certain areas, such as User Experience/Design, Data Analytics, Business Strategy, or Cybersecurity, to refine the suggestions.

METHODOLOGY

The Intelligent Career Path Navigator was developed using CLIPS with a rule-based expert system approach to recommend career paths based on user personality, skills, and interests.

1. Knowledge Acquisition

Knowledge was gathered from analysis of IT career paths and expert insights. The main knowledge elements include user information, personality characteristics (leadership, creativity, analytical thinking), and career interests (software, data, design, security). This knowledge is encoded using CLIPS templates.

2. Rule Formation

Rules are created using IF-THEN logic (`defrule`) to identify personality archetypes, match user profiles with careers, and generate recommendations. Each rule fires automatically when conditions are satisfied.

3. Inference Mechanism

The system uses **forward chaining** where user facts are asserted, the inference engine evaluates rules, matching rules fire, new facts are inferred, and final recommendations are produced.

4. Implementation

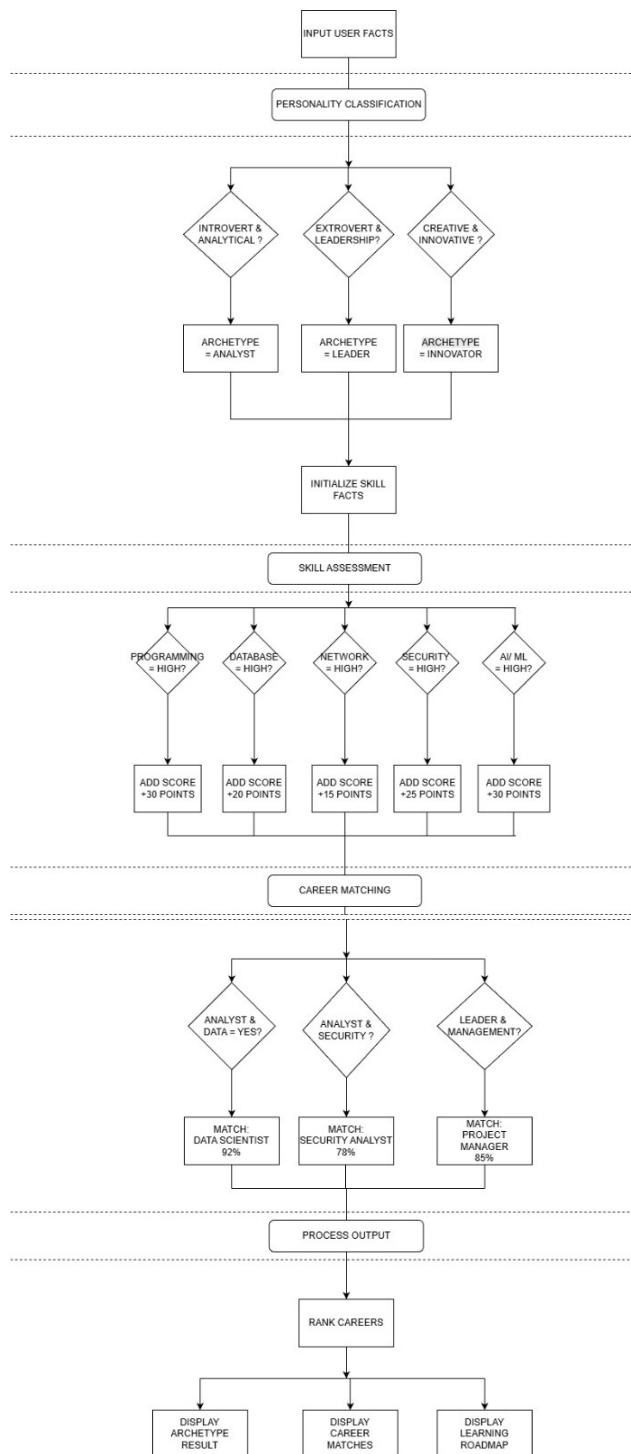
The system is implemented using:

- **(deftemplate)** - structured knowledge representation
- **(deffacts)** - initial system data
- **(defrule)** - reasoning and inference
- **(printout)** - displaying results to user

KNOWLEDGE REPRESENTATION

1. Reasoning Flow

The reasoning process of the system can be represented using the following tree structure:



2. Forward Chaining

The system uses forward chaining as follows:

- i) User data is asserted as facts
- ii) Personality analysis rules are triggered
- iii) A personality archetype fact is inferred
- iv) Career matching rules are evaluated
- v) A career recommendation is produced

This reasoning method follows CLIPS's default inference strategy.

3. Rules Used in the System

The system includes several categories of rules:

- User profile initialization rules
- Personality archetype classification rules
- Career recommendation rules
- Output display rules

All rules are implemented using (defrule) constructs in CLIPS.

CLIPSIDE CODE

Appendix A : CLIPS Source Code

RESULTS & DISCUSSION

```
=====
PERSONALITY PROFILE ANALYSIS
=====

Hello Asynawi!

Your Personality Type: The Collaborator - Team Player
Key Strength: Perfect for team-oriented and client-facing positions

=====
TOP CAREER RECOMMENDATIONS
=====

CAREER: Data Scientist
Match Score: 92% | Confidence: very-high

Why This Career?
Exceptional analytical and mathematical capabilities for data-driven insights

SALARY INSIGHTS (Malaysia):
Entry Level: RM 4,000 - RM 6,500
Senior Level: RM 15,000 - RM 25,000+

MARKET ANALYSIS:
Job Demand: very-high
Remote Work: yes
Growth Rate: 22% - Critical need across all business sectors

=====
CAREER: Full-Stack Software Developer
Match Score: 90% | Confidence: very-high

Why This Career?
Strong coding skills combined with logical problem-solving approach

SALARY INSIGHTS (Malaysia):
Entry Level: RM 3,500 - RM 5,500
Senior Level: RM 12,000 - RM 20,000+

MARKET ANALYSIS:
Job Demand: very-high
Remote Work: yes
Growth Rate: 15% - Consistently high demand across industries

=====
CAREER: AI/Machine Learning Engineer
Match Score: 95% | Confidence: very-high

Why This Career?
Excellent match: Strong mathematical foundation, programming expertise, and passion for data-driven solutions

SALARY INSIGHTS (Malaysia):
Entry Level: RM 5,000 - RM 8,000
Senior Level: RM 18,000 - RM 35,000+

MARKET ANALYSIS:
Job Demand: very-high
Remote Work: yes
Growth Rate: 25% - Fastest growing tech field in Malaysia
```

The system was tested with user "Asynawi". Based on the 16 assessment questions answered, the system produced the following outputs:

Personality Identified: The Collaborator - Team Player, perfect for team-oriented and client-facing positions.

Top 3 Career Recommendations:

1. **AI/Machine Learning Engineer** - 95% match, salary RM 5,000 - RM 35,000+, growth rate 25%
2. **Data Scientist** - 92% match, salary RM 4,000 - RM 25,000+, growth rate 22%
3. **Full-Stack Software Developer** - 90% match, salary RM 3,500 - RM 20,000+, growth rate 15%

Learning Roadmap: Each career includes a 3-phase roadmap with skills, courses, certifications, and project ideas spanning 3-18 months.

Reasoning Behind Outputs

The personality type "Collaborator" was triggered because the user indicated teamwork preference and good communication skills. The career recommendations were generated using **forward chaining inference** by matching user inputs (high math, programming skills, data analytics interest) with rule conditions. AI/ML Engineer scored highest (95%) as it matched all key criteria. The system uses **salience values** to filter and display only the top 3 careers.

CONCLUSION

In conclusion, the Intelligent Career Path Navigator expert system was successfully constructed utilizing the CLIPS programming language and the forward chaining inference approach. The system efficiently collects 16 user inputs, including basic information, technical skills, personality attributes, and job preferences, to create individualized career suggestions. The system uses pattern matching and rule-based reasoning to identify user personality archetypes and proposes the top three most suited IT job pathways, with match rates ranging from 70% to 95%. Furthermore, the system offers full three-phase learning roadmaps that include suggested skills, courses, certifications, and projects for each professional path, as well as Malaysian wage information. This expert system exhibits the actual application of knowledge-based system ideas to real-world career advisory challenges for information technology students.

LINK VIDEO

<https://youtu.be/nyLjWcrluqA>

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APPENDIX

Appendix A: CLIPS Source Code

File Name: career_navigator_updated_v2_2.clp

```
;;;=====
;;; INTELLIGENT CAREER PATH NAVIGATOR v2.1
;;; Enhanced AI-Powered Career Guidance System
;;;=====
;;; Course: BITI2213 Knowledge Based Systems
;;; Institution: UTeM
;;; Semester: 1 2025/2026
;;;=====

;;; =====
;;; ERROR HANDLING FUNCTIONS (NEW ADDITION)
;;; =====

(defun ask-question (?question $?allowed-values)
  (printout t ?question)
  (bind ?answer (read))
  (if (lexemep ?answer) then (bind ?answer (lowercase ?answer)))
  (while (not (member$ ?answer ?allowed-values)) do
    (printout t ">> Invalid input! Please choose from: " ?allowed-
values crlf)
    (printout t ?question)
    (bind ?answer (read))
    (if (lexemep ?answer) then (bind ?answer (lowercase ?answer))))
  ?answer)

(defun ask-number (?question ?min ?max)
  (printout t ?question)
  (bind ?answer (read))
```

```

    (while (or (not (numberp ?answer)) (< ?answer ?min) (> ?answer
?max)) do
      (printout t ">> Invalid input! Please enter a number between "
?min " and " ?max crlf)
      (printout t ?question)
      (bind ?answer (read)))
    ?answer)

;;; =====
;;; TEMPLATES - Data Structures
;;; =====

(deftemplate user-info
  "Basic user demographic information"
  (slot name (type STRING))
  (slot age (type INTEGER))
  (slot education (type SYMBOL) (allowed-values diploma degree
masters phd))
  (slot status (type SYMBOL) (allowed-values student working
unemployed)))

(deftemplate technical-profile
  "Technical skills assessment"
  (slot programming (type SYMBOL) (allowed-values expert
intermediate beginner none))
  (slot mathematics (type SYMBOL) (allowed-values high medium low))
  (slot analytical (type SYMBOL) (allowed-values strong moderate
weak))
  (slot has-coding-experience (type SYMBOL) (allowed-values yes
no))
  (slot learning-pace (type SYMBOL) (allowed-values fast moderate
slow)))

```

```
(deftemplate personality-profile
  "Personality and work style traits"
  (slot problem-approach (type SYMBOL) (allowed-values logical
creative both))
  (slot work-preference (type SYMBOL) (allowed-values independent
teamwork both))
  (slot communication (type SYMBOL) (allowed-values excellent good
average))
  (slot attention-to-detail (type SYMBOL) (allowed-values yes no))
  (slot has-leadership (type SYMBOL) (allowed-values yes no))
  (slot stress-handling (type SYMBOL) (allowed-values high medium
low))
  (slot innovation-level (type SYMBOL) (allowed-values high medium
low)))
```

```
(deftemplate interest-profile
  "Specific career interests"
  (slot design (type SYMBOL) (allowed-values yes no))
  (slot data-analytics (type SYMBOL) (allowed-values yes no))
  (slot business (type SYMBOL) (allowed-values yes no))
  (slot security (type SYMBOL) (allowed-values yes no)))
```

```
(deftemplate personality-archetype
  "User personality classification"
  (slot type (type SYMBOL))
  (slot description (type STRING))
  (slot key-strength (type STRING)))
```

```
(deftemplate career-recommendation
  "Career path recommendation with details"
  (slot title (type STRING))
  (slot match-percentage (type INTEGER))
  (slot confidence (type SYMBOL)))
```

```

(slot reasoning (type STRING))
(slot entry-salary (type STRING))
(slot senior-salary (type STRING))
(slot demand (type SYMBOL))
(slot remote-work (type SYMBOL))
(slot annual-growth (type STRING)))

(deftemplate learning-phase
  "Learning roadmap phases"
  (slot career-title (type STRING))
  (slot phase-number (type INTEGER))
  (slot phase-title (type STRING))
  (slot timeframe (type STRING))
  (slot key-skills (type STRING))
  (slot courses (type STRING))
  (slot certifications (type STRING))
  (slot projects (type STRING)))

;;; =====
;;; INITIAL FACTS
;;; =====

(deffacts system-startup
  "Initialize the system"
  (system-state welcome))

;;; =====
;;; PHASE 1: WELCOME & INTRODUCTION
;;; =====

(defrule welcome-user
  "Display welcome message and system introduction"
  ?state <- (system-state welcome)

```

```

=>
(retract ?state)
(printout t crlf)
(printout t
"===== " crlf)
  (printout t "
" crlf)
  (printout t "  INTELLIGENT CAREER PATH NAVIGATOR v2.0
" crlf)
  (printout t "  AI-Powered Career Guidance System
" crlf)
  (printout t "
" crlf)
  (printout t
"===== " crlf)
    (printout t crlf)
    (printout t "Welcome! This system will help you discover the
perfect" crlf)
    (printout t "IT career path based on your unique profile." crlf)
    (printout t crlf)
    (printout t "What you'll get:" crlf)
    (printout t "  > Personality archetype identification" crlf)
    (printout t "  > Top 3 career recommendations with match scores"
crlf)
    (printout t "  > Personalized 3-phase learning roadmap" crlf)
    (printout t "  > Malaysian salary insights and market data" crlf)
    (printout t crlf)
    (printout t "Let's begin! Please answer the following questions."
crlf)
    (printout t crlf)
    (assert (system-state collect-basic-info)))

;;; =====

```

```

;;; PHASE 2: DATA COLLECTION
;;; =====

(defrule collect-basic-information
  "Gather basic user information"
  ?state <- (system-state collect-basic-info)
  =>
  (retract ?state)
  (printout t "-----"
  ----" crlf)
  (printout t "SECTION 1: BASIC INFORMATION" crlf)
  (printout t "-----"
  ----" crlf)
  (printout t crlf)
  (printout t "What is your name? ")
  (bind ?name (readline))
  (bind ?age (ask-number "How old are you? " 10 99))
  (bind ?edu (ask-question "Education level
(diploma/degree/masters/phd): " diploma degree masters phd))
  (bind ?status (ask-question "Current status
(student/working/unemployed): " student working unemployed))
  (assert (user-info (name ?name) (age ?age) (education ?edu)
(status ?status)))
  (assert (system-state collect-technical)))

(defrule collect-technical-skills
  "Gather technical skills assessment"
  ?state <- (system-state collect-technical)
  (user-info (name ?name))
  =>
  (retract ?state)
  (printout t crlf)
  (printout t "-----"

```

```

----" crlf)
  (printout t "SECTION 2: TECHNICAL SKILLS ASSESSMENT" crlf)
  (printout t "-----")
----" crlf)
  (printout t crlf)
  (printout t "Q1. Programming skills level" crlf)
  (bind ?prog (ask-question "
(expert/intermediate/beginner/none): " expert intermediate beginner
none))
  (printout t crlf)
  (printout t "Q2. Mathematics proficiency" crlf)
  (bind ?math (ask-question "      (high/medium/low): " high medium
low))
  (printout t crlf)
  (printout t "Q3. Analytical thinking ability" crlf)
  (bind ?analytical (ask-question "      (strong/moderate/weak): "
strong moderate weak))
  (printout t crlf)
  (printout t "Q4. Do you have hands-on coding experience?" crlf)
  (bind ?coding (ask-question "      (yes/no): " yes no))
  (printout t crlf)
  (printout t "Q5. How quickly do you learn new technologies?"
crlf)
  (bind ?learning (ask-question "      (fast/moderate/slow): " fast
moderate slow))
  (assert (technical-profile
    (programming ?prog)
    (mathematics ?math)
    (analytical ?analytical)
    (has-coding-experience ?coding)
    (learning-pace ?learning)))
  (assert (system-state collect-personality)))

```



```

(defrule collect-personality-traits
  "Gather personality and work style information"
  ?state <- (system-state collect-personality)
  =>
  (retract ?state)
  (printout t crlf)
  (printout t "-----"
  ----" crlf)
  (printout t "SECTION 3: PERSONALITY & WORK STYLE" crlf)
  (printout t "-----"
  ----" crlf)
  (printout t crlf)
  (printout t "Q6. Problem-solving approach" crlf)
  (bind ?approach (ask-question "      (logical/creative/both): "
logical creative both))
  (printout t crlf)
  (printout t "Q7. Work environment preference" crlf)
  (bind ?work (ask-question "      (independent/teamwork/both): "
independent teamwork both))
  (printout t crlf)
  (printout t "Q8. Communication skills level" crlf)
  (bind ?comm (ask-question "      (excellent/good/average): "
excellent good average))
  (printout t crlf)
  (printout t "Q9. Are you detail-oriented?" crlf)
  (bind ?detail (ask-question "      (yes/no): " yes no))
  (printout t crlf)
  (printout t "Q10. Do you have leadership experience?" crlf)
  (bind ?leader (ask-question "      (yes/no): " yes no))
  (printout t crlf)
  (printout t "Q11. How well do you handle stress?" crlf)
  (bind ?stress (ask-question "      (high/medium/low): " high
medium low))

```

```

(printout t crlf)
(printout t "Q12. Innovation and risk-taking mindset" crlf)
(bind ?innovation (ask-question "      (high/medium/low): " high
medium low))
(assert (personality-profile
  (problem-approach ?approach)
  (work-preference ?work)
  (communication ?comm)
  (attention-to-detail ?detail)
  (has-leadership ?leader)
  (stress-handling ?stress)
  (innovation-level ?innovation)))
(assert (system-state collect-interests)))

(defrule collect-career-interests
  "Gather specific career interests"
  ?state <- (system-state collect-interests)
  =>
  (retract ?state)
  (printout t crlf)
  (printout t "-----"
  ----" crlf)
  (printout t "SECTION 4: CAREER INTERESTS" crlf)
  (printout t "-----"
  ----" crlf)
  (printout t crlf)
  (printout t "Q13. Interested in design and user experience?"
crlf)
  (bind ?design (ask-question "      (yes/no): " yes no))
  (printout t crlf)
  (printout t "Q14. Interested in data analysis and statistics?"
crlf)
  (bind ?data (ask-question "      (yes/no): " yes no))

```

```

(printout t crlf)
(printout t "Q15. Interested in business and strategy?" crlf)
(bind ?business (ask-question "      (yes/no): " yes no))
(printout t crlf)
(printout t "Q16. Interested in cybersecurity?" crlf)
(bind ?security (ask-question "      (yes/no): " yes no))
(assert (interest-profile
  (design ?design)
  (data-analytics ?data)
  (business ?business)
  (security ?security)))
(printout t crlf)
(printout t "Analyzing your profile... Please wait..." crlf)
(printout t crlf)
(assert (system-state analyze-personality)))

;;; =====
;;; PHASE 3: PERSONALITY IDENTIFICATION
;;; =====

(defrule identify-architect-type
  "Identify The Architect personality"
  (system-state analyze-personality)
  (personality-profile (attention-to-detail yes) (problem-approach
logical))
  (technical-profile (analytical strong))
=>
  (assert (personality-archetype
    (type architect)
    (description "The Architect - Systematic & Precise")
    (key-strength "Perfect for roles requiring attention to detail
and structured thinking"))))

```

```
(defrule identify-innovator-type
  "Identify The Innovator personality"
  (system-state analyze-personality)
  (personality-profile (innovation-level high) (problem-approach
creative|both))
  (technical-profile (learning-pace fast|moderate))
  =>
  (assert (personality-archetype
    (type innovator)
    (description "The Innovator - Creative & Bold")
    (key-strength "Excel in cutting-edge technologies and creative
problem-solving"))))
```

```
(defrule identify-strategist-type
  "Identify The Strategist personality"
  (system-state analyze-personality)
  (personality-profile (has-leadership yes) (communication
excellent))
  (technical-profile (analytical strong|moderate))
  =>
  (assert (personality-archetype
    (type strategist)
    (description "The Strategist - Visionary Leader")
    (key-strength "Natural fit for leadership roles and strategic
planning"))))
```

```
(defrule identify-executor-type
  "Identify The Executor personality"
  (system-state analyze-personality)
  (personality-profile (stress-handling high) (work-preference
independent|both))
  (personality-profile (attention-to-detail yes))
  =>
```

```
(assert (personality-archetype
  (type executor)
  (description "The Executor - Reliable & Efficient")
  (key-strength "Thrive in implementation-focused and delivery-
oriented roles"))))
```

```
(defrule identify-collaborator-type
  "Identify The Collaborator personality"
  (system-state analyze-personality)
  (personality-profile (work-preference teamwork|both)
  (communication excellent|good))
  =>
  (assert (personality-archetype
    (type collaborator)
    (description "The Collaborator - Team Player")
    (key-strength "Perfect for team-oriented and client-facing
positions"))))
```

```
(defrule identify-default-type
  "Default personality if no specific match"
  (declare (salience -5))
  (system-state analyze-personality)
  (not (personality-archetype))
  =>
  (assert (personality-archetype
    (type explorer)
    (description "The Explorer - Versatile Learner")
    (key-strength "Adaptable to various roles with eagerness to
learn"))))
```

```
;;; =====
```

```
;;; PHASE 4: CAREER RECOMMENDATIONS
```

```
;;; =====
```

```

(defrule recommend-ai-ml-engineer
  "Recommend AI/ML Engineer career"
  (system-state analyze-personality)
  (technical-profile (mathematics high) (programming
expert|intermediate))
  (interest-profile (data-analytics yes))
  (technical-profile (analytical strong))
  =>
  (assert (career-recommendation
    (title "AI/Machine Learning Engineer")
    (match-percentage 95)
    (confidence very-high)
    (reasoning "Excellent match: Strong mathematical foundation,
programming expertise, and passion for data-driven solutions")
    (entry-salary "RM 5,000 - RM 8,000")
    (senior-salary "RM 18,000 - RM 35,000+")
    (demand very-high)
    (remote-work yes)
    (annual-growth "25% - Fastest growing tech field in
Malaysia")))))

(defrule recommend-full-stack-developer
  "Recommend Full-Stack Developer career"
  (system-state analyze-personality)
  (technical-profile (programming expert|intermediate) (has-coding-
experience yes))
  (personality-profile (problem-approach logical|both))
  =>
  (assert (career-recommendation
    (title "Full-Stack Software Developer")
    (match-percentage 90)
    (confidence very-high)

```

```
(reasoning "Strong coding skills combined with logical
problem-solving approach")
(entry-salary "RM 3,500 - RM 5,500")
(senior-salary "RM 12,000 - RM 20,000+")
(demand very-high)
(remote-work yes)
(annual-growth "15% - Consistently high demand across
industries"))))
```

```
(defrule recommend-data-scientist
  "Recommend Data Scientist career"
  (system-state analyze-personality)
  (technical-profile (mathematics high) (analytical strong))
  (interest-profile (data-analytics yes))
  =>
  (assert (career-recommendation
    (title "Data Scientist")
    (match-percentage 92)
    (confidence very-high)
    (reasoning "Exceptional analytical and mathematical
capabilities for data-driven insights")
    (entry-salary "RM 4,000 - RM 6,500")
    (senior-salary "RM 15,000 - RM 25,000+")
    (demand very-high)
    (remote-work yes)
    (annual-growth "22% - Critical need across all business
sectors"))))
```

```
(defrule recommend-cybersecurity-specialist
  "Recommend Cybersecurity Specialist career"
  (system-state analyze-personality)
  (interest-profile (security yes))
  (personality-profile (attention-to-detail yes))
```

```
(technical-profile (analytical strong|moderate))
=>
(assert (career-recommendation
  (title "Cybersecurity Specialist")
  (match-percentage 88)
  (confidence high)
  (reasoning "Strong security focus with meticulous attention to
detail")
  (entry-salary "RM 4,500 - RM 7,000")
  (senior-salary "RM 15,000 - RM 30,000+")
  (demand very-high)
  (remote-work partial)
  (annual-growth "20% - Critical demand for security
professionals"))))
```

```
(defrule recommend-uiux-designer
  "Recommend UI/UX Designer career"
  (system-state analyze-personality)
  (interest-profile (design yes))
  (personality-profile (problem-approach creative|both))
  (personality-profile (communication excellent|good))
  =>
  (assert (career-recommendation
    (title "UI/UX Designer")
    (match-percentage 85)
    (confidence high)
    (reasoning "Creative mindset with strong design sensibility
and user empathy")
    (entry-salary "RM 3,000 - RM 5,000")
    (senior-salary "RM 10,000 - RM 18,000+")
    (demand high)
    (remote-work yes)
    (annual-growth "13% - Growing importance of user
```



```
experience"))))
```

```
(defrule recommend-devops-engineer
  "Recommend DevOps Engineer career"
  (system-state analyze-personality)
  (technical-profile (programming intermediate|expert))
  (personality-profile (stress-handling high))
  (personality-profile (problem-approach logical|both))
  =>
  (assert (career-recommendation
    (title "DevOps Engineer")
    (match-percentage 87)
    (confidence high)
    (reasoning "Technical proficiency with excellent stress
management for operations")
    (entry-salary "RM 4,500 - RM 7,000")
    (senior-salary "RM 14,000 - RM 22,000+")
    (demand very-high)
    (remote-work yes)
    (annual-growth "18% - Essential for modern software
development"))))
```

```
(defrule recommend-business-analyst
  "Recommend Business Analyst career"
  (system-state analyze-personality)
  (interest-profile (business yes))
  (personality-profile (communication excellent|good))
  (technical-profile (analytical strong|moderate))
  =>
  (assert (career-recommendation
    (title "Business Analyst")
    (match-percentage 80)
    (confidence medium)
```

```
(reasoning "Strong communication skills bridging business and
technology")
(entry-salary "RM 3,500 - RM 5,500")
(senior-salary "RM 8,000 - RM 15,000+")
(demand high)
(remote-work partial)
(annual-growth "11% - Vital link between stakeholders and tech
teams"))))
```

```
(defrule recommend-project-manager
  "Recommend IT Project Manager career"
  (system-state analyze-personality)
  (personality-profile (has-leadership yes) (communication
excellent))
  (personality-profile (stress-handling high|medium))
  =>
  (assert (career-recommendation
    (title "IT Project Manager")
    (match-percentage 83)
    (confidence high)
    (reasoning "Leadership experience with excellent communication
and coordination skills")
    (entry-salary "RM 6,000 - RM 9,000")
    (senior-salary "RM 12,000 - RM 25,000+")
    (demand high)
    (remote-work partial)
    (annual-growth "14% - Always needed for technology
initiatives"))))
```

```
(defrule recommend-default-career
  "Default career recommendation"
  (declare (salience -5))
  (system-state analyze-personality)
```

```

(not (career-recommendation))
=>
(assert (career-recommendation
  (title "IT Support Specialist")
  (match-percentage 70)
  (confidence medium)
  (reasoning "Great entry point to explore various IT domains
and build experience")
  (entry-salary "RM 2,500 - RM 4,000")
  (senior-salary "RM 6,000 - RM 10,000+")
  (demand high)
  (remote-work partial)
  (annual-growth "10% - Foundational role in IT operations"))))

```

```

;;; =====
;;; PHASE 5: LEARNING ROADMAPS
;;; =====

```

```

(defrule create-aiml-roadmap
  "Generate AI/ML Engineer learning roadmap"
  (career-recommendation (title "AI/Machine Learning Engineer"))
  =>
  (assert (learning-phase
    (career-title "AI/Machine Learning Engineer")
    (phase-number 1)
    (phase-title "Foundation Building")
    (timeframe "3-6 months")
    (key-skills "Python programming, Statistics, Linear Algebra,
Calculus basics")
    (courses "CS50 AI, Mathematics for ML (Coursera), Python for
Data Science")
    (certifications "None required at this stage")
    (projects "House price predictor, Image classifier, Sentiment

```

```

analyzer"))))
  (assert (learning-phase
    (career-title "AI/Machine Learning Engineer")
    (phase-number 2)
    (phase-title "Intermediate Development")
    (timeframe "6-12 months")
    (key-skills "TensorFlow, PyTorch, Deep Learning, Natural
Language Processing")
    (courses "Deep Learning Specialization (Andrew Ng), Fast.ai
Practical Deep Learning")
    (certifications "TensorFlow Developer Certificate")
    (projects "Chatbot system, Object detection app,
Recommendation engine"))))
  (assert (learning-phase
    (career-title "AI/Machine Learning Engineer")
    (phase-number 3)
    (phase-title "Advanced Specialization")
    (timeframe "12-18 months")
    (key-skills "Computer Vision, Advanced NLP, MLOps, Model
deployment")
    (courses "Advanced Computer Vision, MLOps Specialization")
    (certifications "AWS ML Specialty, Google Professional ML
Engineer")
    (projects "Production ML pipeline, Research paper
implementation, Kaggle competitions"))))

(defrule create-fullstack-roadmap
  "Generate Full-Stack Developer learning roadmap"
  (career-recommendation (title "Full-Stack Software Developer"))
  =>
  (assert (learning-phase
    (career-title "Full-Stack Software Developer")
    (phase-number 1)

```

```
(phase-title "Frontend Mastery")
(timeframe "2-4 months")
(key-skills "HTML5, CSS3, JavaScript ES6+, React
fundamentals")
(courses "FreeCodeCamp Responsive Web Design, React Official
Tutorial")
(certifications "None required")
(projects "Personal portfolio, Todo app, Weather dashboard"))
(assert (learning-phase
  (career-title "Full-Stack Software Developer")
  (phase-number 2)
  (phase-title "Backend & Databases")
  (timeframe "4-6 months")
  (key-skills "Node.js/Python/Java, SQL & NoSQL, RESTful APIs,
Git/GitHub")
  (courses "Node.js Complete Guide, Database Design Course, API
Development")
  (certifications "Oracle Java SE (optional)")
  (projects "Blog platform, E-commerce API, Social media
clone"))))
(assert (learning-phase
  (career-title "Full-Stack Software Developer")
  (phase-number 3)
  (phase-title "Full-Stack Excellence")
  (timeframe "6-12 months")
  (key-skills "Cloud platforms (AWS/Azure), Docker, CI/CD,
System design")
  (courses "AWS Solutions Architect, Docker & Kubernetes, System
Design Primer")
  (certifications "AWS Certified Developer Associate")
  (projects "Microservices architecture, Cloud-deployed SaaS,
Open-source contributions"))))
```

```

(defrule create-cybersecurity-roadmap
  "Generate Cybersecurity Specialist learning roadmap"
  (career-recommendation (title "Cybersecurity Specialist"))
  =>
  (assert (learning-phase
    (career-title "Cybersecurity Specialist")
    (phase-number 1)
    (phase-title "Security Foundations")
    (timeframe "3-5 months")
    (key-skills "Networking fundamentals, Linux, Windows, Security
principles")
    (courses "CompTIA Security+ prep, Network+, Linux
fundamentals")
    (certifications "CompTIA Security+")
    (projects "Home security lab, Network monitoring,
Vulnerability scanning"))))
  (assert (learning-phase
    (career-title "Cybersecurity Specialist")
    (phase-number 2)
    (phase-title "Ethical Hacking")
    (timeframe "6-9 months")
    (key-skills "Penetration testing, Kali Linux, Web security,
Cryptography")
    (courses "Ethical Hacking Complete Course, Web Application
Security")
    (certifications "CEH (Certified Ethical Hacker)")
    (projects "CTF challenges, Pen testing lab, Security
assessments"))))
  (assert (learning-phase
    (career-title "Cybersecurity Specialist")
    (phase-number 3)
    (phase-title "Advanced Security")
    (timeframe "12-18 months")

```

```
(key-skills "Incident response, Digital forensics, Cloud
security, SIEM tools")
(courses "CISSP preparation, Advanced Forensics, Cloud
Security")
(certifications "CISSP, OSCP (Offensive Security)")
(projects "SOC operations, Incident response playbook, Bug
bounty programs"))))
```

```
(defrule create-uiux-roadmap
  "Generate UI/UX Designer learning roadmap"
  (career-recommendation (title "UI/UX Designer"))
  =>
  (assert (learning-phase
    (career-title "UI/UX Designer")
    (phase-number 1)
    (phase-title "Design Fundamentals")
    (timeframe "2-4 months")
    (key-skills "Design principles, Color theory, Typography,
Figma/Adobe XD")
    (courses "UI Design Fundamentals, Figma Master Class, Design
Theory")
    (certifications "None required")
    (projects "Mobile app mockup, Website redesign, Icon set
design"))))
  (assert (learning-phase
    (career-title "UI/UX Designer")
    (phase-number 2)
    (phase-title "UX Research & Prototyping")
    (timeframe "4-6 months")
    (key-skills "User research, Wireframing, Prototyping,
Usability testing")
    (courses "Google UX Design Certificate, User Research
Methods"))
```

```
(certifications "Google UX Design Professional Certificate")
  (projects "Complete UX case study, Interactive prototype, User
testing report"))))
(assert (learning-phase
  (career-title "UI/UX Designer")
  (phase-number 3)
  (phase-title "Advanced Design Systems")
  (timeframe "6-10 months")
  (key-skills "Design systems, Advanced prototyping, Basic
HTML/CSS, Accessibility")
  (courses "Design Systems Course, Web Accessibility, Motion
Design")
  (certifications "Nielsen Norman Group UX Certification")
  (projects "Complete design system, Portfolio website, Real
client project"))))
```

```
(defrule create-datascientist-roadmap
  "Generate Data Scientist learning roadmap"
  (career-recommendation (title "Data Scientist"))
  =>
  (assert (learning-phase
    (career-title "Data Scientist")
    (phase-number 1)
    (phase-title "Data Fundamentals")
    (timeframe "3-6 months")
    (key-skills "Python/R, Statistics, Probability, SQL, Data
Cleaning")
    (courses "IBM Data Science Professional Cert, Google Data
Analytics")
    (certifications "Certified Associate in Python Programming")
    (projects "Exploratory Data Analysis (EDA), Data Visualization
Dashboard"))))
  (assert (learning-phase
```



```

    (career-title "Data Scientist")
    (phase-number 2)
    (phase-title "Machine Learning Mastery")
    (timeframe "6-12 months")
    (key-skills "Scikit-learn, Feature Engineering, Model
Evaluation, Big Data (Spark)")
    (courses "Machine Learning by Stanford, Spark and Python for
Big Data")
    (certifications "Azure Data Scientist Associate")
    (projects "Customer Churn Prediction, Credit Risk Model,
Recommendation System"))))
(assert (learning-phase
  (career-title "Data Scientist")
  (phase-number 3)
  (phase-title "Advanced Data Science")
  (timeframe "12-18 months")
  (key-skills "Deep Learning, NLP, MLOps, A/B Testing,
Storytelling")
  (courses "Deep Learning Specialization, Storytelling with
Data")
  (certifications "TensorFlow Developer, SAS Certified Data
Scientist")
  (projects "End-to-End ML Pipeline, NLP Sentiment Analysis,
Predictive Maintenance"))))

(defrule create-devops-roadmap
  "Generate DevOps Engineer learning roadmap"
  (career-recommendation (title "DevOps Engineer"))
  =>
  (assert (learning-phase
    (career-title "DevOps Engineer")
    (phase-number 1)
    (phase-title "Infrastructure Basics")

```

```

    (timeframe "3-5 months")
    (key-skills "Linux Admin, Bash Scripting, Networking, Git")
    (courses "Linux Foundation Certified IT Associate, Git &
GitHub Bootcamps")
    (certifications "LPI Linux Essentials")
    (projects "Automated Backup Scripts, Local Web Server
Setup"))))
  (assert (learning-phase
    (career-title "DevOps Engineer")
    (phase-number 2)
    (phase-title "Containerization & CI/CD")
    (timeframe "5-10 months")
    (key-skills "Docker, Kubernetes, Jenkins/GitLab CI,
Terraform")
    (courses "Docker Mastery, Kubernetes Complete Guide")
    (certifications "Docker Certified Associate, CKA
(Kubernetes)"))
    (projects "Containerized Microservices, CI/CD Pipeline"))))
  (assert (learning-phase
    (career-title "DevOps Engineer")
    (phase-number 3)
    (phase-title "Cloud & Advanced DevOps")
    (timeframe "10-18 months")
    (key-skills "AWS/Azure/GCP, Infrastructure as Code, Monitoring
(Prometheus/Grafana)"))
    (courses "AWS DevOps Professional, HashiCorp Terraform
Associate")
    (certifications "AWS DevOps Engineer Professional, Azure
DevOps Engineer Expert")
    (projects "Multi-cloud Deployment, Full Observability
Stack"))))

(defrule create-business-analyst-roadmap

```

```
"Generate Business Analyst learning roadmap"
(career-recommendation (title "Business Analyst"))
=>
(assert (learning-phase
  (career-title "Business Analyst")
  (phase-number 1)
  (phase-title "BA Fundamentals")
  (timeframe "2-4 months")
  (key-skills "Requirements Gathering, Documentation, Process
Mapping, SQL Basics")
  (courses "Business Analysis Fundamentals, SQL for Business
Analysts")
  (certifications "IIBA Entry Certificate in Business Analysis
(ECBA)"))
  (projects "Requirements Document, Process Flow Diagrams"))))
(assert (learning-phase
  (career-title "Business Analyst")
  (phase-number 2)
  (phase-title "Tools & Techniques")
  (timeframe "4-8 months")
  (key-skills "JIRA, Confluence, Data Visualization
(Tableau/Power BI), Agile/Scrum")
  (courses "Agile Business Analysis, Tableau/Power BI Training")
  (certifications "PMI-PBA, Certified Scrum Product Owner
(CSPO)"))
  (projects "Dashboard Creation, User Story Writing, Sprint
Planning"))))
(assert (learning-phase
  (career-title "Business Analyst")
  (phase-number 3)
  (phase-title "Strategic Analysis")
  (timeframe "8-14 months")
  (key-skills "Strategic Planning, Stakeholder Management,
```

```
Change Management")
  (courses "Strategic Business Analysis, Change Management
Certification")
  (certifications "CBAP (Certified Business Analysis
Professional)")
  (projects "Business Case Development, Enterprise Analysis"))))
```

```
(defrule create-project-manager-roadmap
  "Generate IT Project Manager learning roadmap"
  (career-recommendation (title "IT Project Manager"))
  =>
  (assert (learning-phase
    (career-title "IT Project Manager")
    (phase-number 1)
    (phase-title "PM Foundations")
    (timeframe "2-4 months")
    (key-skills "Project Planning, Scheduling, Budgeting, Risk
Management Basics")
    (courses "Google Project Management Certificate, PM
Fundamentals")
    (certifications "CAPM (Certified Associate in Project
Management)")
    (projects "Project Charter, Gantt Chart Creation, Risk
Register"))))
  (assert (learning-phase
    (career-title "IT Project Manager")
    (phase-number 2)
    (phase-title "Agile & Tools")
    (timeframe "4-8 months")
    (key-skills "Agile/Scrum, Kanban, MS Project, JIRA
Administration")
    (courses "Agile Project Management, Scrum Master Training")
    (certifications "CSM (Certified Scrum Master), PMI-ACP"))
```

```

        (projects "Sprint Management, Agile Transformation Plan"))))
(assert (learning-phase
  (career-title "IT Project Manager")
  (phase-number 3)
  (phase-title "Advanced PM")
  (timeframe "8-18 months")
  (key-skills "Program Management, Portfolio Management,
Leadership, Vendor Management")
  (courses "PMP Exam Prep, Leadership for Project Managers")
  (certifications "PMP (Project Management Professional)")
  (projects "Multi-project Portfolio, Vendor RFP Process"))))

(defrule create-it-support-roadmap
  "Generate IT Support Specialist learning roadmap"
  (career-recommendation (title "IT Support Specialist"))
  =>
  (assert (learning-phase
    (career-title "IT Support Specialist")
    (phase-number 1)
    (phase-title "Hardware & OS Basics")
    (timeframe "1-3 months")
    (key-skills "Computer Hardware, Windows 10/11,
Troubleshooting, Customer Service")
    (courses "Google IT Support Professional Certificate")
    (certifications "CompTIA A+")
    (projects "PC Build/Assembly, Windows Installation &
Config"))))
  (assert (learning-phase
    (career-title "IT Support Specialist")
    (phase-number 2)
    (phase-title "Networking & Administration")
    (timeframe "3-6 months")
    (key-skills "Active Directory, Basic Networking (DNS/DHCP),

```

```

Office 365")
    (courses "Windows Server Administration, Networking Basics")
    (certifications "CompTIA Network+, Microsoft 365
Fundamentals")
    (projects "Home Network Setup, User Account Management Lab"))))
(assert (learning-phase
    (career-title "IT Support Specialist")
    (phase-number 3)
    (phase-title "System Administration")
    (timeframe "6-12 months")
    (key-skills "Linux Basics, Scripting (PowerShell),
Virtualization, Security Basics")
    (courses "SysAdmin Training, PowerShell in a Month of
Lunches")
    (certifications "CompTIA Server+, Microsoft Certified: Azure
Fundamentals")
    (projects "Virtual Machine Lab, Automated Backup Scripts"))))

```

```

;;; =====

```

```

;;; PHASE 6: DISPLAY RESULTS

```

```

;;; =====

```

```

(defrule display-personality-result
  "Display identified personality archetype"
  (declare (salience -10))
  ?state <- (system-state analyze-personality)
  (user-info (name ?name))
  (personality-archetype (description ?desc) (key-strength
?strength))
  =>
  (retract ?state)
  (printout t
"===== " crlf)

```

```

(printout t "    PERSONALITY PROFILE ANALYSIS" crlf)
(printout t
"===== " crlf)
(printout t crlf)
(printout t "Hello " ?name "!" crlf)
(printout t crlf)
(printout t "Your Personality Type: " ?desc crlf)
(printout t "Key Strength: " ?strength crlf)
(printout t crlf)
(assert (system-state filter-careers)))

(defrule remove-excess-careers
  (system-state filter-careers)
  ?c <- (career-recommendation (title ?t) (match-percentage
?score))
  (career-recommendation (title ?t1&~?t) (match-percentage ?s1&:(>=
?s1 ?score)))
  (career-recommendation (title ?t2&~?t&~?t1) (match-percentage
?s2&:(>= ?s2 ?score)))
  (career-recommendation (title ?t3&~?t&~?t1&~?t2) (match-
percentage ?s3&:(>= ?s3 ?score)))
  =>
  (retract ?c))

(defrule remove-orphan-roadmaps
  (system-state filter-careers)
  ?roadmap <- (learning-phase (career-title ?title))
  (not (career-recommendation (title ?title)))
  =>
  (retract ?roadmap))

(defrule finish-filtering-transition
  "When filtering is done (only top 3 remain), show the results"

```

```

    (declare (salience -10))
    ?state <- (system-state filter-careers)
    =>
    (retract ?state)
    (assert (system-state display-careers)))

(defrule display-career-recommendations
  "Display all career recommendations"
  ?state <- (system-state display-careers)
  =>
  (retract ?state)
  (printout t
    "===== " crlf)
    (printout t "    TOP CAREER RECOMMENDATIONS" crlf)
    (printout t
    "===== " crlf)
    (printout t crlf)
    (assert (system-state show-careers)))

(defrule show-each-career
  "Show each career recommendation with details"
  (declare (salience -10))
  (system-state show-careers)
  ?rec <- (career-recommendation
    (title ?title)
    (match-percentage ?match)
    (confidence ?conf)
    (reasoning ?reason)
    (entry-salary ?entry)
    (senior-salary ?senior)
    (demand ?demand)
    (remote-work ?remote)
    (annual-growth ?growth))

```



```

=>
(retract ?rec)
(printout t "-----
----" crlf)
(printout t "CAREER: " ?title crlf)
(printout t "-----
----" crlf)
(printout t "Match Score: " ?match "% | Confidence: " ?conf crlf)
(printout t crlf)
(printout t "Why This Career?" crlf)
(printout t ?reason crlf)
(printout t crlf)
(printout t "SALARY INSIGHTS (Malaysia):" crlf)
(printout t "  Entry Level: " ?entry crlf)
(printout t "  Senior Level: " ?senior crlf)
(printout t crlf)
(printout t "MARKET ANALYSIS:" crlf)
(printout t "  Job Demand: " ?demand crlf)
(printout t "  Remote Work: " ?remote crlf)
(printout t "  Growth Rate: " ?growth crlf)
(printout t crlf))

(defrule display-learning-roadmap-header
  "Display learning roadmap section header"
  ?state <- (system-state show-careers)
  (not (career-recommendation))
  =>
  (retract ?state)
  (printout t
"===== crlf)
  (printout t "    YOUR PERSONALIZED LEARNING ROADMAP" crlf)
  (printout t
"===== crlf)

```

```

(printout t crlf)
(assert (system-state show-roadmap)))

(defrule show-each-learning-phase
  "Display each phase of the learning roadmap"
  (system-state show-roadmap)
  ?phase <- (learning-phase
    (career-title ?career)
    (phase-number ?num)
    (phase-title ?title)
    (timeframe ?time)
    (key-skills ?skills)
    (courses ?courses)
    (certifications ?certs)
    (projects ?projects))
  =>
  (retract ?phase)
  (printout t "-----"
  ----" crlf)
  (printout t ?career " - PHASE " ?num ": " ?title crlf)
  (printout t "-----"
  ----" crlf)
  (printout t "Timeframe: " ?time crlf)
  (printout t crlf)
  (printout t "Key Skills to Master:" crlf)
  (printout t " " ?skills crlf)
  (printout t crlf)
  (printout t "Recommended Courses:" crlf)
  (printout t " " ?courses crlf)
  (printout t crlf)
  (printout t "Certifications:" crlf)
  (printout t " " ?certs crlf)
  (printout t crlf)

```

```

(printout t "Project Ideas:" crlf)
(printout t " " ?projects crlf)
(printout t crlf))

(defrule display-final-message
  "Display final motivational message"
  ?state <- (system-state show-roadmap)
  (not (learning-phase))
  (user-info (name ?name))
  =>
  (retract ?state)
  (printout t
    "===== " crlf)
    (printout t "      NEXT STEPS TO SUCCESS" crlf)
    (printout t
      "===== " crlf)
      (printout t crlf)
      (printout t "Congratulations, " ?name "!" crlf)
      (printout t crlf)
      (printout t "ACTION PLAN:" crlf)
      (printout t "1. Choose your top career path from recommendations"
        crlf)
        (printout t "2. Start with Phase 1 of the learning roadmap" crlf)
        (printout t "3. Build projects while learning (learn by doing!)"
          crlf)
          (printout t "4. Connect with professionals on LinkedIn" crlf)
          (printout t "5. Apply for internships after completing Phase 2"
            crlf)
            (printout t crlf)
            (printout t "PRO TIPS:" crlf)
            (printout t "- Focus on one career path at a time" crlf)
            (printout t "- Build a strong portfolio with real projects" crlf)
            (printout t "- Join tech communities and attend meetups" crlf)

```

```
(printout t "- Practice coding daily (consistency is key!)" crlf)
(printout t "- Stay updated with industry trends" crlf)
(printout t crlf)
(printout t
"===== " crlf)
  (printout t "  YOUR JOURNEY TO SUCCESS STARTS NOW!" crlf)
  (printout t
"===== " crlf)
    (printout t crlf)
    (printout t "Thank you for using Intelligent Career Path
Navigator!" crlf)
    (printout t "Good luck on your career journey!" crlf)
    (printout t crlf))
```